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Detection of dislocation motion in atomistic simulations of nanocrystalline materials

Noya Dimanstein Firman[1], **Eliyahu Zvi Engelberg**[2,*] **and Yinon Ashkenazy**[1,*]

[1] Racah Institute of Physics and the Center of Nanoscience and Nanotechnology, The Hebrew University of Jerusalem, Jerusalem 9190401, Israel
[2] Azrieli College of Engineering Jerusalem, Jerusalem 9103501, Israel

E-mail: elien@jce.ac.il and yinona@phys.huji.ac.il

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Abstract

A method for identifying dislocation motion in atomistic simulations is presented. While identifying static and moving dislocations within a single crystal or a combination of such is well established, the method described here is tailored to identify dislocation motion by correlating the displacements of individual atoms. This facilitates the identification of dislocation motion in complex structural arrangements, and allows the specific contribution to plastic deformation, due to dislocation motion, to be separated from that of other mechanisms. The method is applied to test cases in crystals and grain boundaries (GBs), in which irradiation-induced creep (IIC) was induced. It is shown that the method singles out the moving dislocations from among the dislocation forest at GBs, thus identifying the specific reactions driving the distortion at any given time. This enables the study of dislocation processes in the presence of realistic obstacles, and the study of the effects of microstructure on dislocation mobility. As an example of such a study, the method is applied to rule out intragranular slip, and to estimate the contribution of dislocation motion to strain, in a NC undergoing IIC.

Keywords: dislocations, molecular dynamics, graphs, nanocrystals

1. Introduction

Dislocation dynamics often control the plastic response of metals. These dynamics depend on material properties, reactions among dislocations, and their response to ambient conditions such as stress and strain [1]. Therefore, analyzing these dynamics lies at the core of understanding various plastic processes. In most cases, however, direct observation of the

[*] Authors to whom any correspondence should be addressed.

development of dislocation complexes is impossible [2, 3]. One alternative to direct observation is to run molecular dynamics (MD) simulations to study the plastic response of metals. These can provide information regarding the details of the relations between atomistic arrangements and the dislocation processes occurring in metals [4].

At present, it is computationally feasible to run direct MD simulations of nanocrystals (NCs) containing a few tens or hundreds of grains, each of a size of a few tens of nanometers [5]. Such a simulation, with appropriate boundary conditions (periodic or simulating an external load), represents a small part of a macroscopic metal. MD simulations of three-dimensional periodic NC structures have been widely used to study detailed deformation mechanisms under stress [6, 7]. These simulations, together with transmission electron microscope observations, suggest that heterogeneous microstrain evolution in NCs originates in dislocations at the grain boundaries (GBs) [8]. It was shown that, during deformation, partial dislocations are emitted from GBs, leading to the plastic deformation of NCs [9, 10]. Thus, identifying dislocations in GBs, and especially moving dislocations, has become pivotal in the study of the plasticity of NCs.

The classical method used to identify dislocations is Burgers circuit construction, in which paths connecting atoms in the deformed crystal, which is being inspected, are compared to paths in a reference crystalline structure. Algorithms based on this method proved to be extremely robust, but they require the identification, and therefore the existence, of a well-defined reference local crystalline order [11–13]. As such, the effectivity of these algorithms is limited at highly distorted regions, and specifically near GBs and other large-scale defects.

Variation of the underlying structure of the crystal is accounted for in the dislocation extraction algorithm (DXA) [13, 14]. This method can identify dislocations even in large complexes and near GBs, and was used to identify large static distributions of dislocations [13–17]. In some cases, however, the plastic activity of a metal is determined by the motion of only a small number of these dislocations. In such cases, following the changes in the static distribution of all dislocations may be computationally prohibitive, and does not provide enough information to understand the processes driving the plastic activity. This is exacerbated by the fact that the accuracy of tracking all the dislocations in a complex environment is limited. In these cases, to study plastic activity, it is imperative to single out the dislocations driving it, i.e. those which are in motion.

In this paper, we describe a method to identify dislocations that have moved in a crystal. This method utilizes the highly-correlated displacements of individual atoms characterizing dislocation motion. The method, which we call the *common motion grouping* (CMG) method, singles out moving dislocations, does not assume any underlying crystal structure, and is not computationally intensive. It is, therefore, well adapted to the study of plasticity driven by the generation of dislocations at GBs.

2. Description of the CMG method

The CMG algorithm is based on identifying the correlation of the displacements of individual atoms between consecutive time frames. This is done in two steps:

First, atoms whose displacement, relative to their local environment, exceeds $r_{\min}$ are identified. This minimal value is determined to eliminate atoms whose displacement is due to normal thermal Gaussian motion (see appendix A).

Next, the non-thermally displaced atoms are organized as nodes of a binary graph. An edge connects every two atoms that moved in a correlated manner, which is defined as fulfilling two conditions:

1. The two atoms are within distance d of each other, where the value of d is between the nearest-neighbor and the next-nearest-neighbor distances in the lattice (see appendix A).
2. The cosine of the angle between their relative displacements has an absolute value that is greater than $\cos\theta_{\max}$, meaning that their displacements were directionally correlated. For details of how $\cos\theta_{\max}$ is determined, see appendix A.

The atoms are grouped into connected components, i.e. groups of atoms that are all connected, either directly or via other connected nodes [18].

Nonthermal atomic motion can be caused by defects other than dislocations, but will not, in general, involve the correlated motion of several atoms in the same direction. Thus, to differentiate between moving dislocations and atomic motion caused by other defects, only groups that consist of at least N atoms (see appendix A) that moved in a correlated manner are identified as loci of valid moving dislocations.

3. Validation

To validate the CMG algorithm presented in section 2, we examine its performance when applied to the results of the MD simulations of two geometrically simple cases. In both cases, simulations of single fcc Cu grains were performed using `LAMMPS` [19], and employing the widely-used embedded atom method (EAM) potential [20]. In each simulation, two perfect dislocations were introduced. The displacements of atoms due to the motion of these dislocations were then analyzed using the CMG method, and were shown to be consistent with the results of the commonly-used DXA algorithm [14].

In reference simulation A, an fcc lattice was created with dimensions $L_x = 358$ Å, $L_y = 501$ Å, and $L_z = 27$ Å, oriented along the axes shown in figure 1, and with periodic boundaries. Two straight edge dislocations with opposite Burgers vectors parallel to the x axis, $\vec{b}_1 = [\bar{1}10]$ and $\vec{b}_2 = [1\bar{1}0]$, were inserted into the lattice at a distance of $\Delta y = 100$ Å from each other. The dislocations were oriented along the z ($[11\bar{2}]$) axis. Such a dislocation dipole has a lower long-range elastic field, thus allowing simulations to produce convergent results in relatively small systems [21].

Initial relaxation was carried out by simulating the system in an isothermal-isobaric (NPT) ensemble [22] at a temperature of 300 K. In the first 10 ps, a shear stress σ_{xy} was ramped from 0 to 0.2 kbar. Each of the two dislocations initially dissociated in its corresponding glide plane into two Shockley partials, separated by a stacking-fault ribbon, as follows:

$$\begin{aligned}&\frac{1}{2}\left[\bar{1}10\right] \rightarrow \frac{1}{6}\left[\bar{1}2\bar{1}\right] + \frac{1}{6}\left[\bar{2}11\right]\\&\frac{1}{2}\left[1\bar{1}0\right] \rightarrow \frac{1}{6}\left[2\bar{1}\bar{1}\right] + \frac{1}{6}\left[1\bar{2}1\right].\end{aligned} \tag{1}$$

The distance between each two partials fluctuated between 20 and 25 Å. Following this period of relaxation, the stress was held constant at 0.2 kbar for another 10 ps, during which the dissociated dislocations moved away from each other along their glide planes.

The DXA was employed using `Ovito` [23] to detect the dislocations at $t_i = 14$ ps and $t_f = 20$ ps. The CMG method was then applied to detect dislocations that moved between these two time frames. As seen in figure 2, each of the four partial dislocations swept along a surface roughly situated on its xz slip plane. The dislocations detected by the DXA are situated at either side of this surface. The CMG method identified atoms on or close to this surface as proximate atoms with a common direction of motion, thus belonging to one distinct dislocation. Note that

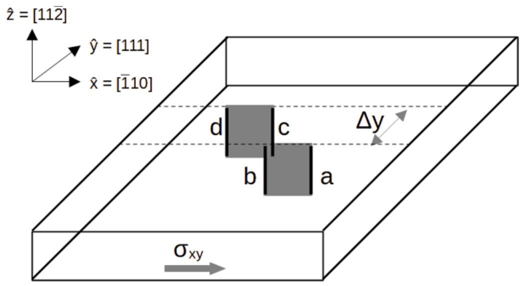

Figure 1. A sketch of the cell of reference simulation A. The lines *a*, *b*, *c*, and *d* indicate the partial Shockley dislocations, with the stacking-fault ribbon marked in gray.

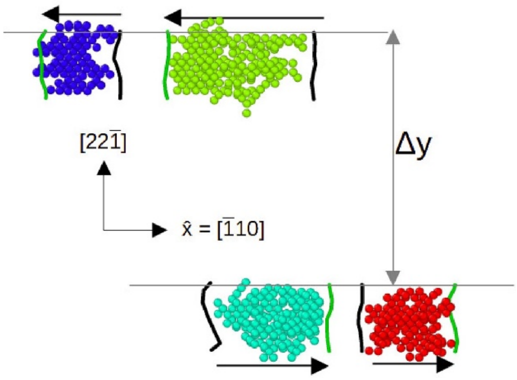

Figure 2. Dislocation map of reference simulation A, looking along the $[001]$ direction. The initial and final positions of the partial dislocations detected by the DXA are shown, with an arrow pointing in the direction of motion. The atoms shown in between were identified by the CMG method as atoms that were displaced due to the motion of the dislocations. This image and others in this manuscript were produced with the help of the `Ovito` program [23].

the method is designed to detect the atoms that changed their position due to the motion of each dislocation, rather than the dislocation line itself.

Similar results were obtained when reference simulation A was repeated at a temperature of 700 K (see appendix B), indicating that the CMG method can be applied across a wide range of temperatures.

In reference simulation B a similar lattice was created, with dimensions $L_x = 27$ Å, $L_y = 501$ Å, and $L_z = 358$ Å, see figure 3. This time, two straight screw dislocations were inserted into the lattice at the same distance as in reference simulation A. The dislocations were oriented along the x axis ($[\bar{1}10]$) with Burger vectors opposite each other, $\vec{b}_3 = [\bar{1}10]$ and $\vec{b}_4 = [1\bar{1}0]$. The simulation was run for the same period of time, with a shear stress σ_{xy} ramped from 0 to 0.2 kbar in the first 10 ps, and then held at 0.2 kbar for another 10 ps. Initially, each of the screw dislocations dissociated into two Shockley partials (equation (1)). The ratio of equilibrium separations of partials from a screw dislocation, vs. partials from an edge dislocation, is 3:7 [24]. Therefore, the partial dislocations stayed closer together than those of reference simulation A, at distances between 10 and 15 Å.

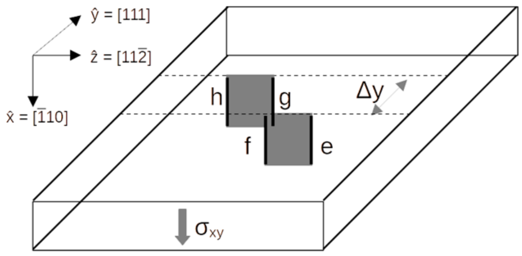

Figure 3. A sketch of the cell of reference simulation B. The lines *e*, *f*, *g*, and *h* indicate the partial Shockley dislocations, with the stacking-fault ribbon marked in gray.

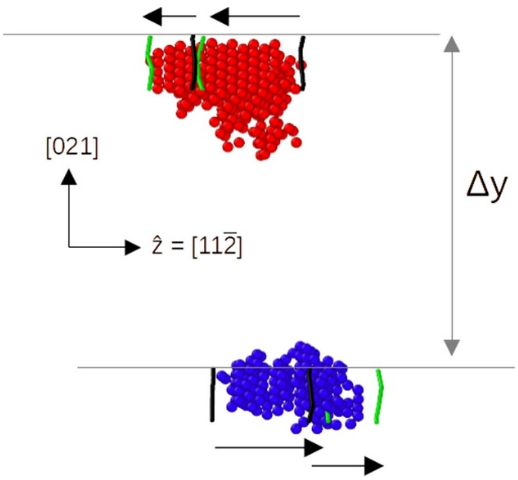

Figure 4. Dislocation map of reference simulation B, looking along the $[\bar{2}0\bar{1}]$ direction. The initial and final positions of the partial dislocations detected by the DXA are shown, with an arrow pointing in the direction of motion. The atoms shown in between were identified by the CMG method as atoms that were displaced due to the motion of the dislocations.

Because of the small dissociation length of the dislocations in reference simulation B, the motion of the Shockley pairs was identified as that of a single complex. The identified displacements, together with the DXA detection of the initial and final partial arrangements at t_i = 1 picosecond and $t_f = 5$ picoseconds respectively, are shown in figure 4. It is seen that the partial dislocations traveled different distances between the two time frames. This is expected, because the width of a dissociated screw dislocation oscillates periodically [21].

Also in contrast with reference simulation A, in which the shifted atoms were close to the slip plane, here atoms farther away from the slip plane were affected, too. This is a result of the different displacement fields of screw and edge dislocations. See for example figure 5, which compares the magnitude of the change in displacement, due to the motion of a screw and edge dislocation in an isotropic elastic medium, calculated from the displacement fields of infinite straight dislocations [24]. Atoms within the black line in figure 5 are displaced by more than 0.7 Å ($= r_{\min}$).

The time frames used for the CMG analysis, in both reference simulations, were chosen so that all four partial dislocations moved between them (i.e. that at least *N* atoms were displaced

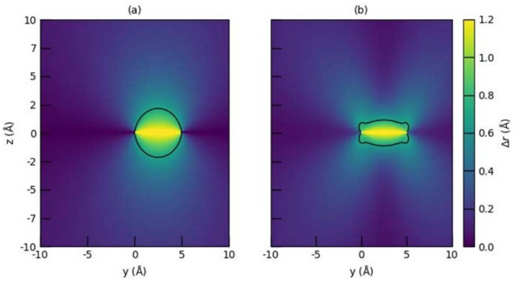

Figure 5. The magnitude of change in displacement in an isotropic elastic medium when a (a) screw and (b) edge dislocation line in the x direction (perpendicular to the plane of the graph) moves from position ($y = 0$, $z = 0$) to ($y = 0$, $z = 5$ Å). The magnitude of the Burgers vector is 2.56 Å, and its direction is along the x axis in (a), and the z axis in (b). In the areas within the black lines, the change in displacement is greater than 0.7 Å ($= r_{\mathrm{min}}$)

due to the motion of each dislocation between the two time frames). On the other hand, they were chosen closely enough so that the motion of the dislocations swept along relatively small areas, and thus it is easy to see the connection between their initial and final positions (found by the DXA), and the groups of atoms that were displaced due to their motion. The comparison of any two time frames, however, yields results that are consistent with the premises of the method.

4. Implementations and performance

With the CMG method validated using the reference simulations of section 3, we examine its performance in a complex realistic configuration of GBs of a NC undergoing creep, for which it is primarily intended. A cubic NC sample of fcc Cu, 224 Å long and composed of seventeen grains, was produced using the method of Voronoi tessellation [25]. GBs were then modified to contain V atoms by switching some of the Cu atoms, using a combined grand canonical Monte-Carlo simulation and MD relaxation, as implemented in `LAMMPS` [26]. By setting an effective reservoir temperature of 300 K and a chemical potential of 2.8 eV, a metastable state with a solute concentration of 0.9% was created. Due to the high miscibility, V atoms were inserted only at GBs, and at this concentration they stabilized the crystal against spontaneous grain growth. The interactions between atoms was calculated using the EAM potential of [27].

To demonstrate the utility of the method, irradiation-induced creep (IIC) was induced by switching Cu atoms at the GBs and at the monolayers adjacent to them [28], i.e. by removing each atom to create a vacancy and reinserting it as an interstitial atom. The CMG method is particularly useful in such a case, because it differentiates between displacements caused by dislocation motion and those due to the irradiation-induced point-defect flux. Seventy atoms were removed and reinserted at a time, after which the sample was allowed to relax for 80 ps. (The insertion of 140 defects at a time was possible because the relaxation process was found to be local.) This was repeated for 1300 cycles to bring the sample to second-stage creep, and then for another 500 cycles, during which measurements were taken. Throughout this process, the sample was subjected to a constant compressive stress of 4 kbar in the [001] direction (z axis) of the simulation box, using the `LAMMPS` NPT command.

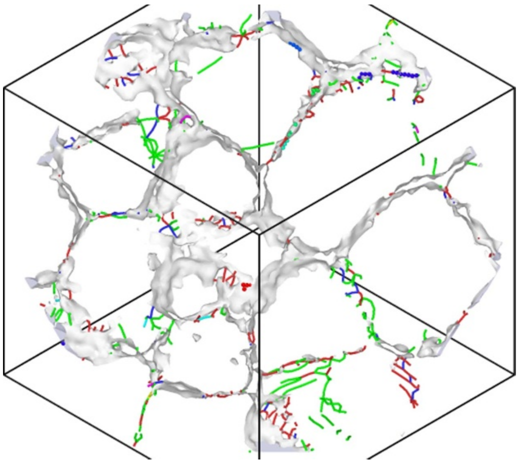

Figure 6. GBs and dislocations in a 10 Å-wide slab along the (111) plane of a Voronoi-tessellated 0.9% solute Cu-V NC, subjected to IIC. A total of 3789 dislocations were detected in the whole NC.

The sample was inspected at the early part of second stage creep, i.e. at $t_f = 106.16$ ns from the beginning of the simulation. At this time frame, the DXA detected 3789 dislocations in the GBs [29]. Figure 6 shows the GBs and dislocations detected by the DXA in a 10 Å-wide slab along the (111) plane of the simulation box.

The CMG method was then applied to detect moving dislocations at the GBs, by analyzing the displacement of atoms during one cycle (of 80 ps). Seventy moving dislocations were identified in the NC by this method. Eight of these moving dislocations were found in the 10 Å-wide slab around the (111) plane, see figure 7 [30]. The isolation of these dislocations allows direct study of the dislocation dynamics in the NC. Furthermore, as most of the moving dislocations were embedded within the highly distorted regions of the GBs, they were identified only by the CMG method and not by the DXA, because the DXA relies on a well-defined local crystalline structure. The motion of these dislocations can be studied in connection with the yield-stress response of the sample in question.

As an example of such an analysis, the moving dislocations detected in the 500 irradiation cycles of second-stage creep were examined. Each moving dislocation displaced a group of atoms. The greatest of these displacements, for each moving dislocation, was saved. In addition, the direction of each dislocation was estimated by finding the direction of the vector connecting the two atoms in the group that were farthest apart. The angles between the dislocation lines and the displacement direction, for the moving dislocations detected by the CMG method at the end of all 500 cycles, were collected into a histogram, shown in figure 8. It is seen that these angles are mostly less than 30°. This is characteristic of dislocations with a dominant screw-type component (see figure 5), including pure and dissociated screw dislocations.

Additionally, the locations of the moving dislocations in the 500 cycles of second-stage creep were analyzed. All 32 183 moving dislocations, with no exception, were found to be within or adjacent to GBs, rather than moving across a grain. We conclude that creep in the simulated sample was affected by GB mobility, with no contribution of intragranular dislocation activity. This finding is in line with previous studies of NC Cu which suggested, based on measured dislocation densities within grains, that GB mobility is the dominant mechanism of plastic deformation when the grains are smaller than ∼10 nm [10, 31–35].

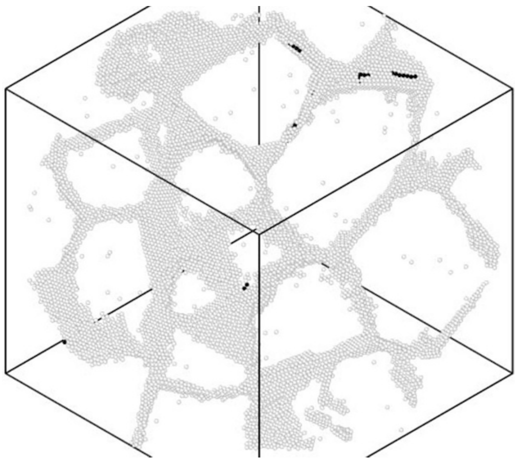

Figure 7. GBs and moving dislocations in a 10 Å wide slab along the (111) plane of a Voronoi-tessellated 0.9% solute Cu-V NC, subjected to IIC. The black atoms belong to the eight moving dislocations within this slab, out of a total of 70 moving dislocations in the whole NC.

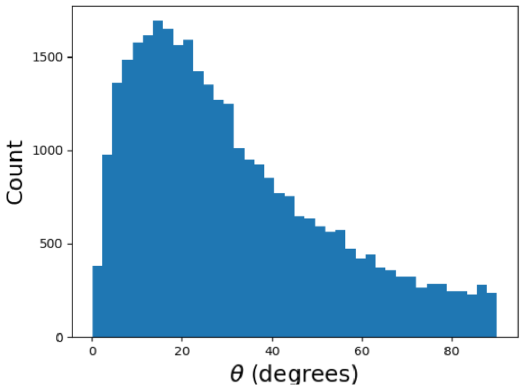

Figure 8. Histogram of the angle between the estimated dislocation orientation and the direction of greatest displacement in the dislocation, in all dislocations detected by the CMG algorithm in 500 cycles of IIC. The histogram contains 40 bins.

Finally, the contribution of the moving dislocations to the linear strain of the sample was estimated, by summing the atoms within the groups, and comparing the result to the change in the dimensions of the simulation box. It was found that, after the first 250 cycles of second-stage creep, there was a strong correlation between the displacements and the box dimensions. Figure 9 shows the cumulative sum of atom displacements as a function of the elongation of the simulation box in the z-axis direction, along which compressive stress was applied. The Pearson correlation coefficient of the two data sets is 0.946, and the p-value under the null hypothesis is 3×10^{-123}. Initial local relaxation can lead to the formation of back stress, displacing atoms in directions other than the compressive axis. This can explain the fact that the correlation begins only after the first 250 cycles. The complete mechanism, by which dislocation motion within GBs is translated into global deformation, can be studied further.

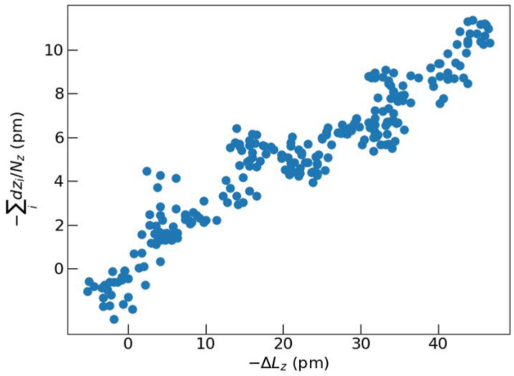

Figure 9. Cumulative sum of atom displacements as a function of the elongation of a Voronoi-tessellated 0.9% solute Cu-V NC, subjected to IIC, along the [001] (z axis) direction. The sum is divided by the number of atoms in each plane perpendicular to the z-axis direction.

5. Limitations

As shown in the previous sections, the CMG method can be used to detect moving dislocations both within grains and at GBs. Its ability to single out the moving dislocations at GBs is significant, because it allows the analysis of NC plasticity near GBs. The method works, furthermore, without assuming an underlying crystal structure, thus making it flexible and robust. In this section we list and discuss a few of the limitations of the CMG method. Some of these limitations are inherent, and others remain because the method is still in development.

Slow-moving dislocations. By design, the CMG method only detects moving dislocations. If a dislocation moves slowly, the displacement of atoms between two close time frames might be less than the threshold of detection. The sampling rate of the CMG method must correspond, therefore, to the typical speed of dislocations in the system under inspection.

Dislocation forests. As seen in reference simulation B of section 3 (in which a dissociated screw dislocation was identified as one screw dislocation rather than two partials), the CMG method identifies the motion of proximate dislocations as that of one complex dislocation. This is an inherent feature of the method since it analyzes the displacements of groups of atoms. If the motion of a group of atoms is correlated, they will all belong to the same connected component of the CMG graph. Conversely, if at least N atoms are affected by one dislocation and not others, they will belong to a separate connected component, and will thus be identified as a separate dislocation.

Burgers vector. The CMG method represents atoms as nodes of a binary graph, connected by edges if they are nearby and move in the same (or opposite) direction. If the path on the graph between two atoms is long, their direction of motion might be very different, even though they belong to the same dislocation, see figure 5. The magnitudes of the displacements of atoms, too, vary within one dislocation. Therefore, the detection of a dislocation does not automatically yield its Burgers vector, as is the case, for example, in the DXA. Such analysis is in fact possible, but requires a more detailed study of the specific displacement vectors, and is left for future extension.

Dislocation lines. The CMG method identifies groups of atoms whose positions shift due to moving dislocations, rather than tracing the dislocation lines themselves. In other words, the method will not pinpoint the regions of the crystal lattice around which Burgers circuits fail to close. This can be an advantage when studying the effect of multiple dislocations on the lattice and grain structure of a sample, but might be a disadvantage when studying the velocities of isolated dislocations [28, 36–38]. The method can be developed to support such studies by improving the accuracy of the identification of the edges of the areas swept by the moving dislocations and incorporating improved static structure criteria. In the current implementation, the velocity of a dislocation might be estimated by calculating the rate of propagation of the center of mass of the atoms that are shifted by its motion.

Misidentification of deformations as dislocations. In metastable scenarios with high energies, the ensuing dynamics may involve correlated motion of chunks of matter, which is not driven by the motion of dislocations. The CMG method does not differentiate between chunks of matter moving together and moving dislocations. This type of scenario can include the initial steps of some simulations, in which the system has not yet relaxed into a low-energy state, and situations in which failure is incipient.

6. Conclusions

Detailed local study of plasticity in NCs and other structures, containing large populations of defects and dislocations, requires the identification of a small number of moving dislocations in complex environments. The CMG method, introduced here, facilitates such studies by identifying correlated displacements due to moving dislocations. This is done by comparing atomic positions at consecutive time frames. The method was applied to quantify and characterize displacements due to moving dislocations in a NC undergoing creep, and to estimate the contribution of these dislocations to the NC strain. Using this method, it was shown that all dislocation activity was identified within the GB, with no significant contribution due to intragranular activity.

Although the method now compares two specific time frames, it can be extended to examine multiple time steps. This will allow the calculation of properties such as dislocation drag and velocity, and better filtering of specific dislocation mechanisms. Also, tools can be developed to determine the Burgers vectors of the detected dislocations, to facilitate the study of the types of dislocations that drive plasticity in various settings.

In addition to the study of plasticity in NCs, the CMG method can be used in other settings in conjunction with static structural methods, to discern among dislocations that moved and those that remained stationary. This can include situations in which sessile dislocations become mobile, as a result of reactions induced by stress or radiation. The identification of the onset of dislocation-based displacements, in such settings, can provide information regarding the activation energies and volumes of different types of yield mechanisms.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://doi.org/10.5281/zenodo.10356082.

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Appendix A. Determining the parameters of the algorithm

Table 1 lists the parameters used in the current implementation of the CMG algorithm, and their values in the simulations presented in this manuscript (except in the simulation described in appendix B). In this appendix, we detail how these values were derived.

$r_{\min}$ is used to identify atoms that changed their position between two time frames under comparison. Its value was calculated using the atomic positions at $t_i = 106.08$ ns and $t_f = 106.16$ ns from the beginning of the GB simulation of section 4. The magnitudes of the atomic displacements between these time frames were arranged in a histogram and fitted to a χ distribution with three degrees of freedom, see figure 10, under the assumption that thermal displacements lead to a normal distribution in each spatial dimension. Thus, $r_{\min} = 0.7$ Å, where the displacements noticeably diverge from the χ_3 distribution, was chosen as the threshold of non-Gaussian motion.

Although the value of $r_{\min}$ can be determined separately for every simulated sample (at every point of time), its calculation yields similar values for a given material and temperature. Thus, the value yielded by the calculation above was used in all the simulations described in this paper, except in that of appendix B.

The next parameter in the CMG algorithm, d, is used to determine the correlation length of atoms moving due to one dislocation. We expect dislocations to affect contiguous groups of atoms, at adjacent positions along the crystal lattice. Therefore, to determine d, we examine a histogram of distances among pairs of atoms in the simulated NC at $t_f = 106.16$ ps, produced by `Ovito`'s pair distribution function [23]. d is any value between the two leftmost peaks of the histogram. In our case, as shown in figure 11, it can be $d = 3$ Å. (Once again, choosing a different time frame from the NC simulation, or from the reference simulations, yields similar results.) In general, in fairly ordered crystals, d should have a value between the distances among nearest and next-nearest neighbors.

To estimate the next parameter, $\cos\theta_{\max}$, we consider that two atoms, displaced by the passage of a moving dislocation, should be displaced in the same direction. Typically, for the fcc crystals simulated in this work, the magnitude of this displacement will be equal to the Burgers vector of a partial dislocation, i.e. $b = a/\sqrt{6} = 1.48$ Å, where a is the lattice constant in Cu. Thermal motion, however, can cause the directions of the displacements to diverge. This will be most pronounced when the thermal motions of the atoms are in opposite directions, perpendicular to the motion due to the passage of the moving dislocation. We estimated the maximum displacement of thermal motion as $r_{\min}$, so the maximum angle between the directions of motion of two atoms will be $\theta_{\max} = \arctan(r_{\min}/b)$. Therefore, the minimum cosine of the angle between the displacement vectors will be $\cos\theta_{\max} = 0.9$.

The reasoning applied to estimate $\cos\theta_{\max}$ assumes an fcc lattice structure with fairly consistent atomic spacing, and where partial Shockley dislocations are common. Similar reasoning can be developed in other settings, too, as long as interatomic distances are greater than the typical thermal displacement of atoms. Along these lines, the CMG method may be extended to highly complex systems such as high-entropy alloys.

The last parameter of the CMG algorithm is N, the minimum amount of moving atoms that indicates the passage of a moving dislocation. A large value for this parameter would lead to the misdetection of small moving dislocations. A small value would lead to the false identification of moving point defects as moving dislocations, since there is a motion of atoms around point defects due to relaxation. Under the assumption that not all of the nearest neighbors around a point defect will move in a common direction, we chose $N = 12$ as the optimal value for the fcc lattices in this work.

Table 1. List of parameters of the CMG algorithm.

Parameter	Description	Value
$r_{\min}$	Threshold of non-Gaussian displacement	0.7 Å
d	Maximum distance of nearby atoms	3 Å
$\cos\theta_{\max}$	Minimum cosine of angle for common motion	0.9
N	Minimum number of atoms in group	12

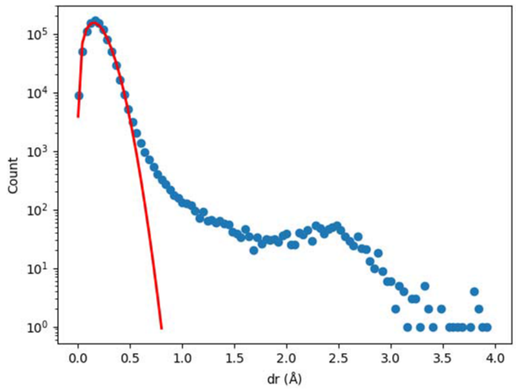

Figure 10. Frequency histogram of the magnitude of the displacement of atoms during 80 ps, in a simulation of a Cu NC with 0.9% V solute (circles), fitted to a χ_3 distribution (line).

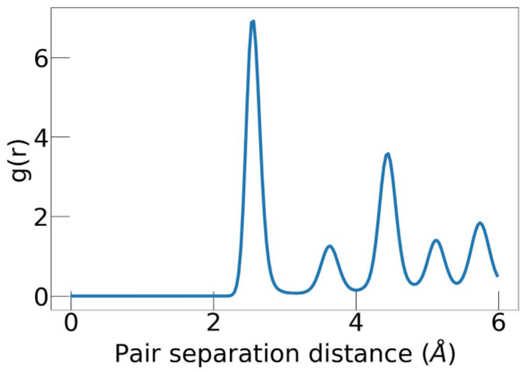

Figure 11. Pair distribution function of one time frame in a simulation of a Cu NC with 0.9% V solute.

Appendix B. Reference simulation A at 700 K

To test the validity of the CMG method at higher temperatures, reference simulation A (see section 3) was rerun at a temperature of 700 K. First, using the procedure described in appendix A and comparing the atomic positions at two different time frames of the simulation, it was found that $r_{\min} = 0.9$ Å and, therefore, $\cos\theta_{\max} = 0.85$. The other two parameters of the algorithm (see table 1) were not changed. Next, the CMG algorithm was run. Figure 12 shows the atoms identified as belonging to moving dislocations, when comparing the atomic

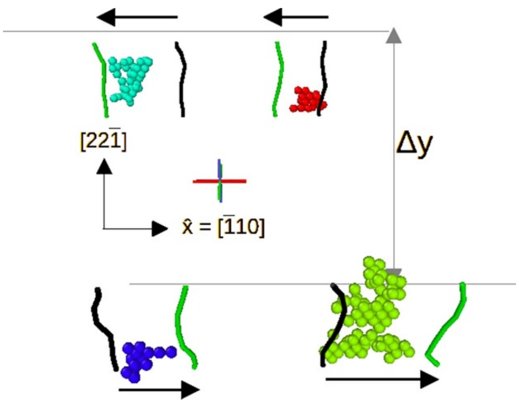

Figure 12. Dislocation map of reference simulation A rerun at a temperature of 700 K, looking along the [001] direction. The initial and final positions of the partial dislocations detected by the DXA are shown, with an arrow pointing in the direction of motion. The atoms shown in between were identified by the CMG method as atoms that were displaced due to the motion of the dislocations.

positions at $t_i = 11$ ps and $t_f = 20$ ps from the beginning of the simulation. As in section 3, these time frames were chosen due to the visual clarity of the results of the DXA and the CMG method.

The CMG method singles out atoms that have shifted by more than r_{min} between time frames, to examine if their motion is due to moving dislocations. r_{min} itself is determined statistically per sample. As the temperature increases, thus widening the χ_3 distribution of the thermal motion of atoms, the concept of singling out atoms by comparing two frames will become less effective. It can be expected to break down when there is a statistically significant likelihood for atoms to be shifted by a magnitude greater than that of the Burgers vector of the dislocations of interest. We see, however, that at 700 K in the scenarios in this study, this concept continues to be effective. At even higher temperatures, the algorithm might be modified by inspecting the mean coordinates of the atoms of the sample over the time period of interest, thus reducing the effect of Brownian motion on the CMG method.

ORCID iDs

Eliyahu Zvi Engelberg https://orcid.org/0000-0002-1591-4273
Yinon Ashkenazy https://orcid.org/0000-0002-7976-4534

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